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The Editors-in-Chief
Journal of Cheminformatics
BioMed Central
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Dear Editors,

We are pleased to submit our manuscript, **“Topological and tensor-network representations resolve chemical paradoxes in African antimalarial natural products”**, for consideration as an Original Research Article in the *Journal of Cheminformatics*.

This work introduces three molecular representations—Topological Fingerprint (TFP), Tensor Network Embedding (TNE), and Quantum Kernel Score (QKS)—and benchmarks them against classical fingerprints on a library of 19,836 African natural-product-derived antimalarial candidates. Persistent homology resolves a scaffold paradox (92.6% Tanimoto novelty with 69.3% scaffold recovery) by separating H_0 (connectivity) from H_1 (ring topology). TNE achieves a $6.1\times$ real-atom mean compression while retaining binding-relevant information. Quantum kernels, evaluated under a rigorously controlled protocol, are statistically indistinguishable from a properly tuned classical RBF baseline at all sample sizes (6-qubit QKS AUC 0.823 versus RBF 0.829 at $n = 19,849$, $p = 0.060$).

We report honest negative results: on the corrected canonical benchmark, classical ECFP4 remains the strongest single descriptor for primary screening (AUC 0.948). The hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888, significantly below ECFP4 ($p < 0.0001$) yet above every standalone quantum-inspired descriptor, with QKS as the principal positive contributor (ablation $\Delta = -0.040$). Cross-paper analysis with a companion molecular-dynamics study shows that H_1 count correlates with resistance-resilience scores, although the association is mediated by molecular size.

Recent systematic reviews of quantum computing in bioinformatics and drug discovery have called for rigorous empirical evaluation of near-term quantum methods on real pharmaceutical datasets. Our reproducible benchmark answers this call with a fully specified protocol, public code and transparent reporting of both positive and negative findings. We believe the work aligns closely with the scope of the *Journal of Cheminformatics*: reusable open-source molecular descriptors, a carefully controlled benchmark, and an honest assessment of when quantum-inspired methods do not outperform classical baselines.

Code and currently available benchmark artefacts are accessible to reviewers in the public repository (https://github.com/NanaEngo/Malaria_codesV2) under the MIT licence. The Zenodo DOI <https://doi.org/10.5281/zenodo.19608875> is reserved; the upload remains pending and will be completed before final publication. This work is original, is not under consideration elsewhere,

and all authors have approved the submission.

We thank you for your consideration.

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Suggested reviewers:

- Prof. Bing Wang (topological methods in drug discovery)
- Dr. Guillaume Godin (RDKit, natural product cheminformatics)
- Prof. Gisbert Schneider (generative molecular design)